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P R O J E C T D O C U M E N T A T I O N

CareerPilot AI

AI-Powered Career & Upskilling Advisor

From a **job advert** to a **personalised learning plan** in five connected AI modules: **Skill Extraction, ESCO Linking, Career Transitions, Learner Model, and Course Recommender**.

Item	Detail
Type	Hackathon MVP v0.1
Shared vocabulary	ESCO v1.2.1 (skills and occupations)
Repository	github.com/Mawa009/Career_Upskilling_Advisor_Plan
Modules	M1 Extraction, M2 Linking, M3 Transitions, M4 Learner Model, M5 Recommender

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1. Project at a Glance

CareerPilot AI reads a **job advert**, finds the **skills** the job needs, checks **what the learner already knows**, and recommends the **fewest hours of courses** that close the gap.

> ■ **CORE FORMULAS**
> **Priority(skill) = importance x (1 - mastery)**
> **Score(course) = sum of priority of covered skills / max(course hours, 5)**
> **P(next job B | job A) = N(A to B) / sum N(A to any job)**

Module	What it does	Main model	Main dataset
M1 Skill Extraction	Job advert to skill phrases	**JobBERT** (SkillSpan) + ESCO trie + tech lexicon	SkillSpan, ESCO
M2 ESCO Linking	Phrase to official ESCO skill	**all-MiniLM-L6-v2** + exact match	ESCO v1.2.1
M3 Career Transitions	Job to likely next jobs	**Markov chain** (1st and 2nd order)	JobHop v2
M4 Learner Model	Quiz to mastery per skill	**Bayesian Knowledge Tracing**	60 hand-written MCQs
M5 Recommender	Gaps to ranked courses	**Greedy diverse scorer** + MiniLM tagging	Coursera dataset

2. System Architecture

Modules talk to each other through small **JSON / CSV files**, all keyed by **ESCO v1.2.1 skill IDs**. Any module can be replaced without breaking the others.

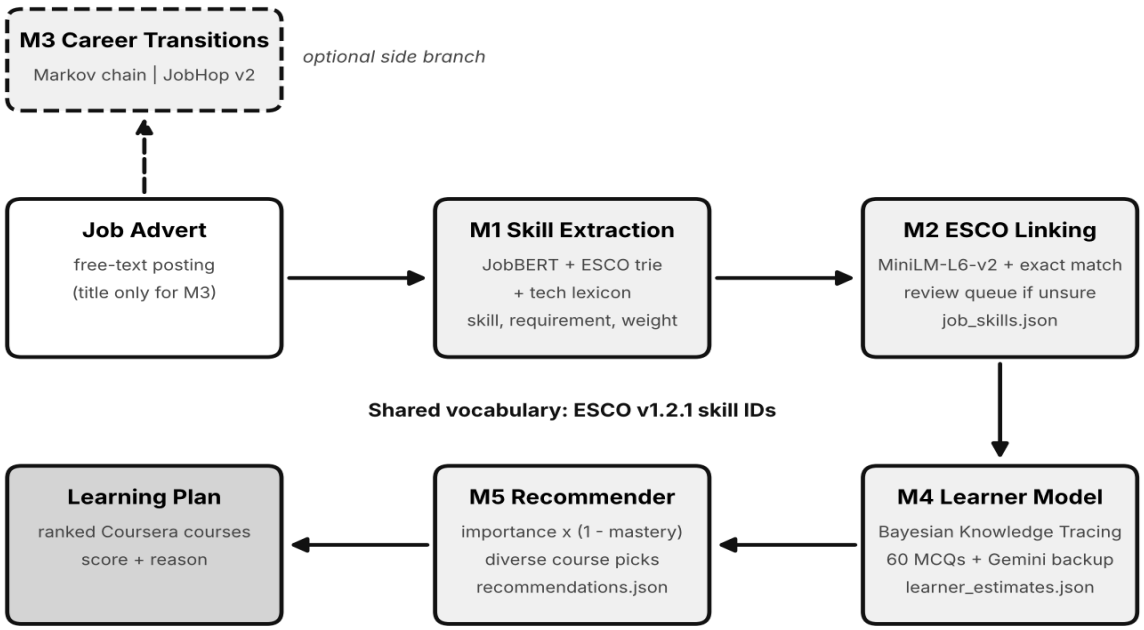


Figure 1. End-to-end pipeline. M3 is an optional side branch.

Hand-off	File	Fields
M1 to M2	`member1_output.csv`	skill, count, sources, confidence, requirement, importance_hint
M2 to M4 / M5	`job_skills.json`	esco_id, label, importance
M4 to M5	`learner_estimates.json`	ESCO URI to mastery (0 to 1), plus `estimate_sources.json`
M5 output	`recommendations.json`	course_id, score, reason, Coursera link

3. Module 1: Skill Extraction

> ■ **GOAL**
> Read a **job advert** and return ranked **skill phrases**, each with a **requirement level** and an **importance weight**.

Models used

- **JobBERT skill tagger**: `jjzha/jobbert_skill_extraction` (BERT token classification, B/I/O tags, trained on **SkillSpan**)
- **JobBERT knowledge tagger**: `jjzha/jobbert_knowledge_extraction` (tools, languages, frameworks)
- **ESCO phrase trie**: dictionary lookup over all ESCO skill labels
- **Curated tech lexicon**: 163 hand-made entries (Python, SQL, Power BI, TensorFlow, ...)

How it works

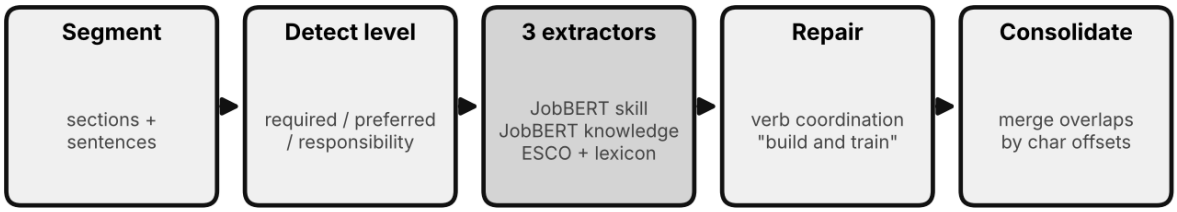


Figure 2. M1 pipeline

Requirement level	importance_hint
required	1.0
responsibility	0.8
unknown	0.7
preferred	0.5
company boilerplate	0.3

Datasets: SkillSpan (training data of JobBERT), ESCO v1.2.1, LinkedIn-50 (50 ad excerpts, 5 job families), 5-ad synthetic dev set.

Validation: ablation study and per-category scores (Section 10), plus **36 automated tests** (all pass).

> ■ **KEY RESULT**
> Full pipeline reaches **F1 0.78** (partial match) on LinkedIn-50; the honest range for unseen ads is **0.73 to 0.78**. The **knowledge model** gives the biggest jump (0.37 to 0.73).

4. Module 2: ESCO Linking

> ■ **GOAL**
> Turn each M1 phrase into an **official ESCO skill ID** so every module speaks the same language. Doubtful matches go to a **human review queue** instead of being guessed.

Models used

- **Exact matcher**: normalised label / alias lookup (`EXACT_LABEL`, `EXACT_ALIAS`, `AMBIGUOUS_EXACT_ALIAS`)
- **Semantic matcher**: `sentence-transformers/all-MiniLM-L6-v2` embeddings + cosine similarity (cached for speed)
- **TF-IDF** (bigram) kept as an older backup version

Safety gates

Gate	Value	Meaning
`MIN_SCORE`	**0.84**	Minimum similarity to accept a link
`MIN_MARGIN`	**0.05**	Best match must beat the 2nd best by this much
`TOP_K`	3	Candidates shown to a reviewer
`BLOCKED_ALIAS_PAIRS`	list	Known bad links, e.g. TensorFlow to computer vision

Dataset: ESCO v1.2.1 `skills_en.csv` (about 13,900 released skills).

> ■ KEY RESULT
> On the sample advert, **3 skills were accepted** (Python, SQL, Power BI) and **6 were held for review**. This is the **safe-over-wrong** design: no confident wrong links reach the learner.

5. Module 3: Career Transitions

> ■ GOAL
> Given a current occupation, predict the **top-k likely next occupations**. An **optional side branch** that does not feed M4 or M5.

Models used

- **First-order Markov chain**: transition counts with a minimum-support filter
- **Second-order Markov model**: uses the previous job as extra context; **backs off** to first-order when data is thin (`min_context_support=5`, `smoothing=10`)
- **Popularity baseline**: always recommends the most common jobs

Dataset: JobHop v2 (Hugging Face `aida-ugent/JobHop`, Flemish resumes mapped to ESCO occupations, train/val/test, **CC BY 4.0**). One career path per person, ordered by start quarter; ambiguous same-quarter starts are skipped.

Validation: leave-last-transition-out on held-out people, measured by **Recall@5, MRR and coverage**, compared against the popularity baseline; **3 unit tests** pass.

> ▲ STATUS
> Training and evaluation scripts are complete and trained models are saved. **Final metric values were not stored in the repository**, so none are quoted here.

6. Module 4: Learner Model

> ■ GOAL
> Quiz the learner on the job's ESCO skills and produce a **mastery score (0 to 1) per skill**.

Models used

- **Bayesian Knowledge Tracing (BKT)**: own implementation; defaults prior 0.30, guess 0.20, slip 0.10 (**not fitted**)
- **Smoothed-accuracy baseline** for comparison
- **Gemini** (optional): drafts questions for skills with no question bank
- **Self-report fallback**: learner rates a skill when no quiz exists

Dataset: 60 hand-written MCQs, 12 each for **SQL, Python, Data visualisation, Statistics, Project management** (20 easy, 20 medium, 20 advanced).

Service: FastAPI backend (`/health`, `/v1/skills/coverage`, `/v1/assessments/start`, `/v1/assessments/submit`, `/v1/estimates/from-history`) and a **Streamlit** quiz app. Every score is labelled by source: `experimental_bkt`, `self_report` or `llm_generated_quiz_experimental_bkt`.

Validation: 44 automated tests pass (Gemini mocked).

> ▲ **IMPORTANT**
> BKT parameters are **defaults, not fitted**. A score is an **estimate**, not a certification. Skills with no questions show as **not assessed**, never as zero.

7. Module 5: Gap Analysis and Course Recommender

> ■ **GOAL**
> Combine **job demand** and **learner mastery** into skill gaps, then recommend the **fewest course hours** that cover them.

Models and algorithms used

- **Gap scorer**: priority = importance x (1 - mastery)
- **Course scorer**: priority covered / max(hours, 5)
- **Prerequisite and difficulty filters**: advanced courses hidden until average mastery is 0.5
- **Greedy diverse picker**: after each pick, covered skills drop out so the list never repeats
- **Course tagging**: keywords OR `all-MiniLM-L6-v2` similarity per skill

Dataset: Coursera course dataset (`azrai99/coursera-course-dataset`): **6,645 courses**; **450** tagged to five skills.

Validation (Experiment C): four strategies on two simulated learners, **Ayesha** and **Bilal**, measured by **gap coverage** and **total hours**.

Strategy	Ayesha (coverage / hours)	Bilal (coverage / hours)
Popularity (enrolment)	0.24 / 618 h	0.41 / 618 h
Demand-only	1.00 / 20 h	1.00 / 20 h
Diverse (ours)	**1.00 / 11 h**	**1.00 / 10 h**

> ■ **KEY RESULT**
> The diverse recommender covers **all gaps in about 10 to 11 hours**, versus **618 hours** for a popularity list. Caveat: learners are **simulated** and coverage is partly circular (the method optimises the same objective).

8. User Interface

The **CareerPilot AI web app** is a single workspace with six pages. The status badge at the top right shows the **Member 4 API is online**, so the assessment pages run on the live learner-model service.

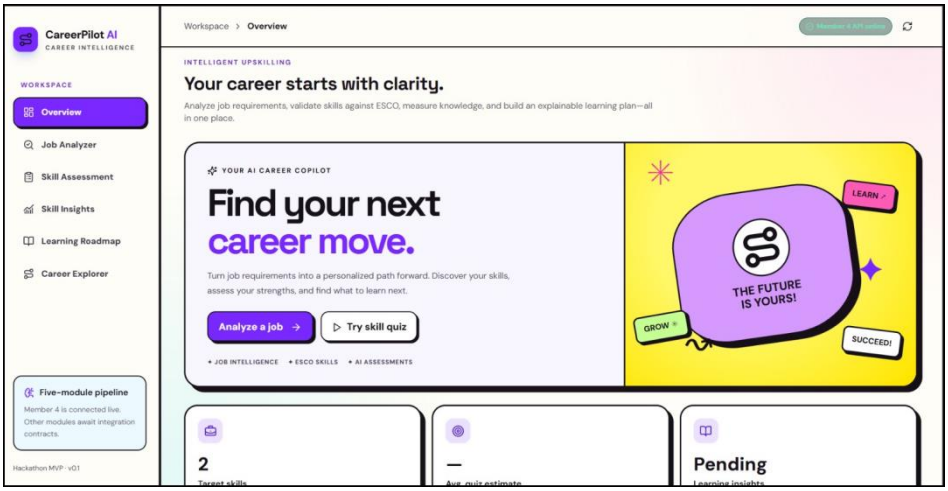


Figure 3. CareerPilot AI workspace: Overview page

Page	Purpose
Overview	Landing page with **Analyze a job** and **Try skill quiz** buttons, plus summary cards (target skills, average quiz estimate, learning insights)
Job Analyzer	Paste a job advert to get required skills (M1 + M2)
Skill Assessment	Take the quiz for the job's skills (M4)
Skill Insights	See mastery per skill and the gaps (M4 + M5)
Learning Roadmap	Ranked course plan (M5)
Career Explorer	Next-job suggestions (M3)

> **▲ INTEGRATION STATUS**
> The sidebar shows **"Member 4 is connected live. Other modules await integration contracts."** The full chain M1 to M5 has been demonstrated in the integration notebook; wiring every page of the web app to the other modules is the next step.

9. Models and Datasets Used

Models

Stage	Model	Type	Pretrained or fitted
M1	**JobBERT skill + knowledge**	BERT token classifier	Pretrained (SkillSpan)
M1	**ESCO trie + lexicon (163)**	Dictionary matcher	Hand-built
M2	**all-MiniLM-L6-v2**	Sentence embeddings + cosine	Pretrained
M2	**Exact matcher, TF-IDF**	Rules / backup	n/a
M3	**Markov 1st and 2nd order**	Transition counts + backoff	Fitted on JobHop v2
M4	**Bayesian Knowledge Tracing**	Hidden Markov mastery	Default parameters
M4	**Gemini (optional)**	LLM question generator	External API
M5	**MiniLM + greedy diverse scorer**	Tagging and ranking	Pretrained / rule-based

Datasets

Dataset	Used by	Notes
ESCO v1.2.1	M1, M2, M3, M4	Skills and occupations; shared vocabulary
SkillSpan	M1 (via JobBERT)	Annotated skill spans in job ads (NAACL 2022)
LinkedIn-50	M1 validation	50 ad excerpts, 5 job families; draft labels
JobHop v2	M3	Career histories with ESCO codes; CC BY 4.0
60 MCQ bank	M4	12 questions for each of 5 skills; needs expert review
Coursera dataset	M5	6,645 courses; 450 tagged to 5 skills
Simulated learners	M4, M5 tests	Ayesha and Bilal

10. How We Validated the System

■ **THE VALIDATION STORY**

1. **Trust the foundations:** JobBERT, ESCO, MiniLM and JobHop are published, peer-reviewed resources.
2. **Test the riskiest step first:** if the advert is read wrongly, everything after it fails, so M1 got a real test.
3. **Prefer safe over wrong:** M2 links only confident matches and sends the rest to review.
4. **Compare with baselines:** M3 against popularity, M5 against popularity and demand-only.
5. **Test the whole chain:** one real advert runs end to end, backed by **83 automated tests** (36 + 3 + 44).
6. **Be honest about limits:** learners are simulated, BKT values are defaults, labels are draft.

Module	Validation method	Status
M1	Ablation on LinkedIn-50, precision / recall / F1, per-category scores, 36 tests	Done
M2	End-to-end run and review queue inspection	No accuracy figure yet
M3	Leave-last-transition-out, Recall@5, MRR, 3 tests	Scripts done; numbers not stored
M4	44 pytest tests, mocked Gemini	No fitted metrics
M5	Experiment C: 4 strategies, 2 simulated learners	Done at small scale
Whole system	Integration notebook, one advert through M1 to M5	Runs end to end

M1 ablation on LinkedIn-50 (partial match)

Treat **0.78 as an upper bound** and **0.73 as the safer figure**: labels are **draft**, ads are **short excerpts**, and the lexicon was written while looking at these ads.

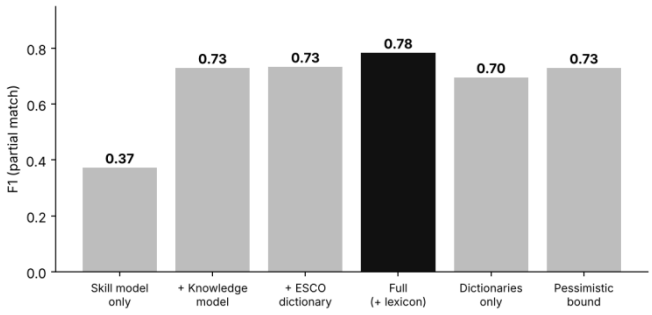


Figure 4. Each component improves F1; the full system is highlighted.

Configuration	Precision	Recall	F1
Skill model only	0.627	0.266	**0.373**

Configuration	Precision	Recall	F1
+ Knowledge model	0.733	0.728	**0.730**
+ ESCO dictionary	0.734	0.734	**0.734**
Full (+ tech lexicon)	0.754	0.817	**0.784**
Dictionaries only (no BERT)	0.926	0.557	**0.696**
Lexicon minus gold terms (pessimistic)	0.725	0.734	**0.730**

By job family (full, F1): Software Development **0.93**, ML/AI **0.86**, Data Analytics **0.82**, Digital Marketing **0.70**, Project Management **0.56** (weakest).

11. Limitations and Responsible AI

Area	Limitation	Next step
M1	Draft labels, short excerpts, lexicon built from same ads	Review labels; test on unseen full ads
M2	No accuracy measured; 6 of 9 sample skills needed review	Hand-check about 100 phrases
M3	Flemish data only; result numbers not saved	Re-run final test and store metrics
M4	BKT not fitted; 60 questions; Gemini questions not fact-checked; in-memory sessions	Expert review; fit on real learners
M5	Only 5 skills tagged; catalogue noise; Recall@5 / NDCG@5 not computed	Tag by ESCO ID; evaluate on real users
System	M4 and M5 tested on simulated learners	Pilot with real users

Responsible AI

- **Human in the loop:** uncertain skill links go to a review queue.
- **Transparency:** every score is labelled with how it was produced.
- **Not a certificate:** mastery values are estimates, not proof of job readiness.
- **Privacy and security:** no stored learner accounts; API keys only via environment variables.

12. Research Papers and Why We Used Them

Paper / resource	Why we used it	Status
SkillSpan (NAACL 2022)	Source of the **JobBERT** models and labelled job-ad spans behind M1	**Used**
ESCO taxonomy	One **common skill and occupation vocabulary** for all modules	**Used**
JobHop (IEEE BigData 2025)	**Real career histories mapped to ESCO** to fit the M3 model	**Used**
Course recommenders must consider the job market (SIGIR 2024)	Justifies **demand-aware** course ranking in M5	**Idea used**
pyBKT (2021)	Reference for the **Bayesian Knowledge Tracing** formulation in M4	Foundation only
all-MiniLM-L6-v2 (Sentence-Transformers)	Small, fast **semantic matcher** for M2 and M5 tagging	**Used**
simpleKT (ICLR 2023)	Deep knowledge-tracing alternative for M4	Cited, not built

Paper / resource	Why we used it	Status
NNOSE (EACL 2024)	Retrieval to boost skill extraction	Planned, not built
Tabiya ESCO classifier (2025)	Planned ESCO linker; gated, so replaced by MiniLM	Not used
CAREER (2022)	Optional deep career-path model	Cited, not used

Links: SkillSpan aclanthology.org/2022.naacl-main.366 | JobHop arxiv.org/abs/2505.07653 | SIGIR 2024 arxiv.org/abs/2404.10876 | pyBKT arxiv.org/abs/2105.00385 | simpleKT arxiv.org/abs/2302.06881 | NNOSE aclanthology.org/2024.eacl-long.35 | Tabiya arxiv.org/abs/2512.03195 | CAREER arxiv.org/abs/2202.08370 | ESCO esco.ec.europa.eu